



SQL ARCHITECTURE

Automatic Production Line Analytics

Version 1.0
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Automatic Production Line Analytics



OBJECTIVES

- Clean raw manufacturing data
- Standardize business logic
- Create reusable SQL views
- Build optimized reporting dataset
- Support Power BI analytical model



DESIGN PRINCIPLES

- Layered architecture
- Reusable SQL views
- Business logic separation
- Performance optimization
- Easy maintenance



RAW DATA



SQL VIEWS



ANALYTICAL
DATASET



POWER BI



SQL ARCHITECTURE

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SQL VIEW ARCHITECTURE

VIEW	PURPOSE	INPUT	OUTPUT / GRAIN
 vw_Production_Detail	Cleans and aggregates Production Data. Merges Quality, Production Data and Article Base	Production Data, Quality Data and Article Base	Date+Shift+Line+Product+Tool
 vw_ProductionPlanMain	Unpivots and standardizes production planning data	Production Planning (header + details)	Date+Shift+Line+Product (Planned Data)
 vw_OEEAutoLine	Combines Production, Quality, Planning, Parameters data and calculates all KPIs	vw_Production_Detail, vw_ProductionPlanMain, ProductionLineParameter	Date+Shift+Line+KPI DataSet

BENEFITS



REUSABLE



MODULAR



EASY
MAINTENANCE



FAST
REPORTING